CENG381 Stochastic Processes

Chapman-Kolmogorov Equations and Diagonalization - Answer Key 1

Q1 Solution

a) C-K equation and P^2(E,E):

The Chapman-Kolmogorov equation for n=m=1 states:

P^2(E, E) = sum over k in {E, W} of P^1(E, k) * P^1(k, E)
= P(E,E)*P(E,E) + P(E,W)*P(W,E)
= 0.7 * 0.7 + 0.3 * 0.4
= 0.49 + 0.12
= 0.61

b) Eigenvalue lam2 = 1 - p - q:

lam2 = 1 - 0.3 - 0.4 = 0.3

Verify lam2 = 0.3 is an eigenvalue via det(P - lam2*I) = 0:

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P - 0.3*I = [ 0.7-0.3   0.3  ] = [ 0.4   0.3 ]
            [ 0.4      0.6-0.3]   [ 0.4   0.3 ]
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det(P - 0.3*I) = 0.4*0.3 - 0.3*0.4 = 0.12 - 0.12 = 0 (OK)

So lam2 = 0.3 is indeed an eigenvalue of P.

c) Closed-form P^n(E,E) via diagonalization:

With p = 0.3, q = 0.4, p+q = 0.7, lam2 = 0.3:

P^n(E, E) = q/(p+q) + lam2^n * p/(p+q)
= 0.4/0.7 + (0.3)^n * 0.3/0.7
= 4/7 + (0.3)^n * 3/7

Check for n=2:
P^2(E,E) = 4/7 + (0.09)*(3/7) = 4/7 + 0.27/7 = 4.27/7 ~ 0.610 (OK)

As n -> infinity: (0.3)^n -> 0, so P^n(E,E) -> 4/7 ~ 0.571.
This is exactly pi(E) = q/(p+q), the stationary probability of E.
The chain converges to its stationary distribution regardless of
where it starts, at an exponential rate governed by |lam2| = 0.3.

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Q2 Solution

a) Row sums of Q:

Row 0: -1/3 + 1/4 + 1/12

= -4/12 + 3/12 + 1/12 = 0/12 = 0 (OK)

Row 1: 1/3 - 1/2 + 1/6

= 2/6 - 3/6 + 1/6 = 0/6 = 0 (OK)

Row 2: 1/6 + 1/3 - 1/2

= 1/6 + 2/6 - 3/6 = 0/6 = 0 (OK)

b) Embedded chain P (divide off-diagonal entries by total rate out):

Total rate out of state 0: |Q(0,0)| = 1/3

P(0, 1) = Q(0,1) / (1/3) = (1/4) / (1/3) = 3/4

P(0, 2) = Q(0,2) / (1/3) = (1/12)/ (1/3) = 1/4

P(0, 0) = 0 (no self-loops in the jump chain)

Total rate out of state 1: |Q(1,1)| = 1/2

P(1, 0) = Q(1,0) / (1/2) = (1/3) / (1/2) = 2/3

P(1, 2) = Q(1,2) / (1/2) = (1/6) / (1/2) = 1/3

Total rate out of state 2: |Q(2,2)| = 1/2

P(2, 0) = Q(2,0) / (1/2) = (1/6) / (1/2) = 1/3

P(2, 1) = Q(2,1) / (1/2) = (1/3) / (1/2) = 2/3

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Embedded chain P:
          0      1      2
   0  [   0     3/4    1/4  ]
   1  [ 2/3      0     1/3  ]
   2  [ 1/3    2/3      0   ]
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c) P^2(0, 2) using C-K:

P^2(0, 2) = sum over k in {0,1,2} of P(0,k) * P(k,2)

= P(0,0)*P(0,2) + P(0,1)*P(1,2) + P(0,2)*P(2,2)

= 0 * (1/4) + (3/4)*(1/3) + (1/4)*0

= 0 + 1/4 + 0

= 1/4 = 0.25

So starting from Off (state 0), there is a 25% chance of being
in High power (state 2) after exactly 2 jumps of the embedded chain.

Q3 Solution

a) Row sum verification:

For any i >= 0:

P(i, 0) + P(i, i+1) = 1/(i+1) + i/(i+1) = (1+i)/(i+1) = 1 (OK)

Every row sums to exactly 1, confirming P is a valid stochastic matrix.

b) P^2(0,0) and discussion:

At state i = 0:

P(0, 0) = 1/(0+1) = 1 and P(0, 0+1) = P(0,1) = 0/(0+1) = 0

So from state 0, the chain STAYS at state 0 with probability 1.
State 0 is therefore an ABSORBING state under this definition.

P^2(0, 0) = P(0,0)*P(0,0) + P(0,1)*P(1,0) = 1*1 + 0*(1/2) = 1.

Interpretation: once the chain reaches state 0 it never leaves,
making state 0 absorbing (positive recurrent).

c) Modified chain -- is state 0 recurrent?

Modified: P(0,1)=1 (state 0 always moves to state 1).
For i >= 1: P(i,0) = 1/(i+1), P(i,i+1) = i/(i+1).

Starting from state 1, the probability of EVER returning to state 0 is:

h(1) = P(1,0) + P(1,2)*h(2)

= 1/2 + (1/2)*h(2)

h(i) = P(i,0) + P(i,i+1)*h(i+1) = 1/(i+1) + i/(i+1)*h(i+1)

We compute the probability of absorption at 0 starting from each state.
Using the result for infinite random walks with decreasing return probs:

P(return to 0 from state 1) = sum_{n=1}^{inf} P(first reach 0 at step n)

A simpler approach: the probability of going from state k directly to 0 is 1/(k+1). The probability of NEVER returning to 0, starting from 1, satisfies a system. For this chain it can be shown that:

sum_{k=1}^{inf} 1/(k*(k+1)) using telescoping:
1/(k*(k+1)) = 1/k - 1/(k+1)

sum_{k=1}^{N} [1/k - 1/(k+1)] = 1 - 1/(N+1) -> 1 as N -> inf

This sum equals 1, which indicates that starting from state 1, the chain returns to state 0 with probability 1.
=> State 0 is RECURRENT in the modified chain.